

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The traffic logs show DNS requests sent over UDP to the DNS server at 203.0.113.2, followed by ICMP responses stating “udp port 53 unreachable.” Port 53 is used for DNS queries and responses, which help translate domain names into IP addresses. The affected application-layer protocol was DNS, which used UDP for the requests shown in the capture. This failure prevented the browser from resolving the website’s domain name and accessing www.yummyrecipesforme.com.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The captured traffic begins at 13:24:32. Several customers reported being unable to access www.yummyrecipesforme.com and receiving a “destination port unreachable” error. The cybersecurity team reproduced the problem by attempting to load the website, then used the network analysis tool tcpdump to investigate.

The logs show repeated DNS requests to 203.0.113.2, followed by ICMP responses reporting that UDP port 53 was unreachable. A possible cause is that the DNS service was not running or listening on UDP port 53. A firewall configuration blocking access to the port could also explain the issue. Another possible cause is a denial-of-service attack disrupting the DNS service, although the captured traffic does not confirm an attack.

Next steps include checking the firewall configuration and asking the DNS server administrator to verify that the DNS service is running and accepting requests on UDP port 53. Reviewing server logs and checking for unusually high traffic volumes would help investigate a possible denial-of-service attack. Further investigation is needed to determine the root cause and restore access to the website.